RW-6102

503101

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

BIOMOLECULES AND BIOTECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Define ELISA.

2. Bioenergetics.

3. Monosaccharides.

4. Aminoacids.

5. Poly nucleotides.

6. EMB Agar.

7. Feeder cell line.

8. LB media.

9. DNA ligation.

10. Cosmid.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Draw the structures of sulphur containing amino acids with examples and give their importance.

(*Or*)

(b) Illustrate with an example, the differences between plant and animal cell.

12. (a) Highlight the scope and importance of biotechnology.

(*Or*)

(b) Discuss on Classical Vs. Modern concepts of Biotechnology.

13. (a) Give the structural organization of nucleic acids and also state their clinical significances.

(*Or*)

(b) What are micro and macro nutrients ? Give their effects on the biological systems.

14. (a) Explain the different plating techniques employed in microbiology.

(*Or*)

(b) Comment on the process of micro propagation.

15. (a) What are the restriction enzymes ? Give their significant role in molecular biology.

(*Or*)

(b) Define cDNA. How are they constructed ?

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Write an essay on disaccharides.

17. Derive the Michiaelis-Menten formalism.

18. Give the lipid classification with its impact on biological system.

19. Comment on :

    (a) Cell culture method.

    (b) Plant cell culture method.

    (c) Bacterial culture method.

    (d) Viral culture method.

20. Explain the strategies in Gel electrophoresis with its application.

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RW-6103

503102

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

NANOSCIENCE AND MATERIALS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Explain the term Amorphous Structure.

2. State Bragg's Law.

3. What are extended defects in Solid Structure ?

4. What is meant by dislocation in Solids ?

5. Define Protein Crystal.

6. What are nanocrystals ?

7. What is meant by exciton ?

8. What is meant by Zero-dimensional nanostructure ?

9. Give examples of nanostructured materials ?

10. What is DNA Computing ?

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Differentiate Crystalline and Amorphous materials.

(*Or*)

(b) Explain the working Principle of Transmission Electron Microscope..

12. (a) Explain the nucleation process.

(*Or*)

(b) Discuss the role of defects on the properties of Solids.

13. (a) Compare the properties of microscopic and macroscopic Crystals.

(*Or*)

(b) Explain top-down and bottom-up approaches.

14. (a) Discuss about two-dimensional nanostructured materials.

(*Or*)

(b) Explain about metals in nanostructured materials.

15. (a) Explain the preparation techniques of ZnO nanotubes.

(*Or*)

(b) Explain about nanorods and nanowires.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail about X-ray diffraction.

17. Explain the methods involved in the growth of Single Crystal.

18. (a) How the nucleation rate influences the size of the Crystal ?

    (b) How the surface to Volume ratio changes from micron to nano crystal ?

19. Expalin in detail about one-dimensional nanostructured materials.

20. (a) Explain in detail single and Multiwall Carbon nanotubes.

    (b) Explain any one technique to prepare carbon nanotube.

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RW-6104

503103

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

BIOELECTRONICS—I

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. What is transducer ?

2. What are electronic noses ?

3. What is the relations describing the relationship between the current and potential ?

4. What is coulostatic response ?

5. Define molecular arrays.

6. What is unit assembly ?

7. Give an example for biomolecular switches.

8. What is non-polarisable electrode ?

9. Why three electrode system is used in conventional electrochemistry ?

10. What are different bio-sensing techniques ?

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the applications of noble metals in biosensors ?

(*Or*)

(b) Write biosensor applications in medicinal science.

12. (a) Briefly describe field effect transistors for pH measurement.

(*Or*)

(b) Differentiate the features between the cyclic and steady state voltammetric behaviour of $K_3(Fe(CN)_6$.

13. (a) Explain the recent advancements in molecular wires and switches.

(*Or*)

(b) Write briefly the properties of DNA and its applications in molecular electronics.

14. (a) Describe electrical double layer theories with suitable diagrams.

(*Or*)

(b) Describe the assembly of photonic biomolecular memory stores.

15. (a) Explain the different types of transducers used for biosensors.

(*Or*)

(b) Write commercial aspects of biosensors.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail electronic noses and tongues and their applications in biosensors.

17. Describe ISFET with suitable example for glucose sensing.

18. Explain in detail the applications of Biosensors in industrial and environmental applications.

19. Explain the properties of DNA and its applications in molecular electronics.

20. What are electrochemical transducers and explain their applications in different types of biosensors with suitable examples of target analytes.

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RW-6105 | 503701

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

Elective—NUMERICAL MATHEMATICS AND PROGRAMMING

(CBCS—2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Calculators may be allowed.

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Give the purpose of each one of the six unary operators.

2. Explain the switch - case statement.

3. Why is it sometimes desirable to pass a pointer to a function as an argument ?

4. Define a structure declaration.

5. Explain normalized floating point representation.

6. How is secant method different from Newton-Raphson method ?

7. How is pivoting done ?

8. How does Gauss - Jordan method differ from Gauss method ?

9. Use the trapezoidal rule to evaluate $\int_1^4 f(x)\,dx$ given :

x	:	1	2	3	4
$f(x)$	:	2.105	2.808	3.614	4.604

10. How is Euler method improved in modified Euler method ?

Part - B

(5 × 5 = 25)

Answer **all** questions choosing *either* (a) *or* (b).

11. (a) Give the purpose of the six assignment operators. How can multiple assignment be written in C ? In what older will the assignments be carried out ?

(*Or*)

(b) Explain with examples, the construction of loop using

(i) do - while statment.

(ii) for statement.

12. (a) How is a function defined and accessed in a program ? What are function prototypes ? What is their purpose ?

(*Or*)

(b) What is the relationship between an array name and a pointer ? How is an array name interpreted when it appears as an argument to a function ? How can the array be declared within the function ?

13. (a) Obtain Newton Raphson iteration formula for $f(x) = 0$. Use it to find the cubic root of 7 with an accuracy of three decimal places.

(*Or*)

(b) Explain the successive approximation method of solving $f(x) = 0$.

14. (a) Explain Gauss Seidel method of solving a set of simultaneous equations.

(*Or*)

(b) Explain the method of least squares to fit a curve.

15. (a) Obtain Simpson's $\frac{1}{3}$ rule for integration.

(*Or*)

(b) Explain the second order Runge - Kutta method of solving differential equation.

Part - C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Write a loop using "if-else" statement to generate every third integer, beginning with $i = 2$ and continuing for all integers upto 100. Calculate the sum of those integers that are evenly divisible by 5.

17. Write a C program that will enter a line of text store it in an array and then display it backwards.

18. Perform five iterations of the bisection method to obtain the smallest positive root of the equation $f(x) = x^3 - 5x + 1 = 0$.

19. Apply Lagrange's formula to interpolate $f(1.5)$ using the following values of the function $f(x)$

x	:	1.00	1.20	1.40	1.60	1.80	2.00
$f(x)$	:	0.2420	0.1942	0.1497	0.1109	0.079	0.0540

20. Solve, by using the Runge - Kutta method, the equation

$$\frac{dy}{dx} = \frac{1}{(x+y)} \text{ for } x = 0.5\ (0.5)\ 2, \text{ given } y\ (0) = 1.$$

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RW-6106

503301

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

BIOSENSORS—I

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section - A (10 × 2 = 20)

Answer **all** questions.

1. What is the role of transducer in biosensor ?

2. Write few examples of medical applications of biosensor.

3. Explain enzyme catalysis with activation diagram.

4. What is the advantage of Multiple enzyme biosensors over Single enzyme biosensors ?

5. What are the applications of biomembrane ?

6. What is an ion channel ?

7. Write a note on electrochemical potential.

8. Distinguish Potentiometric and Amperometric techniques.

9. What are the applications of semiconductors ?

10. Explain chemiluminescence with example.

Section - B (5 × 5 = 25)

Answer **all** questions.

11. (a) What are the advantages and limitations of glucose biosensors ?

(*Or*)

(b) Describe in detail the biosensors in medical diagnosis of diabetes and cancer.

12. (a) Write a note on Industrial glucose monitoring.

(*Or*)

(b) Discuss mechanistic aspects of single and multiple enzymatic biosensors.

13. (a) Explain in detail the importance of ion channels in biosensing.

(*Or*)

(b) Discuss and distinguish Potentiometric and Amperometric membrane electrodes.

14. (a) Briefly discuss different types of transducers with suitable examples.

(*Or*)

(b) Discuss the applications of optical biosensors with suitable examples.

15. (a) Write a note on piezoelectric biosensors.

(*Or*)

(b) Explain how to construct the molecular electronics based transducers.

Section - C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the advantages and limitations of biosensors with an example and methods to overcome the limitations.

17. Describe different immunological techniques and their applications.

18. Discuss the general properties and applications of biomembranes in biosensing.

19. Compare and contrast the optical and electrical biosensors with suitable examples.

20. Discuss the recent advancements in chemiluminescence based biosensing.

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RW-6107

503302

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

NANO BIOTECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Transgenic animal.

2. Hyaluronate.

3. Polar molecules.

4. Mitochondria.

5. Carbon tube.

6. Nanomagnetic materials.

7. Affimetric method.

8. Actively driven transport.

9. Nanocrystals.

10. CdS nanoparticles.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Write short notes on membrane array.

(*Or*)

(b) Discuss in detail about the lipid chips.

12. (a) Describe in detail about confocal microscopy.

(*Or*)

(b) Write a note on 2 D gel electrophoresis.

13. (a) Briefly describe the Dip-Pen lithography.

(*Or*)

(b) Give a detail account on immunosensor.

14. (a) Discuss in detail about the docking.

(*Or*)

(b) Explain in detail about nanomedicine.

15. (a) Give a detail account on chip based DNA detection assay.

(*Or*)

(b) Discuss in detail about the microbial nanoparticle production.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Give a detail account on Lab-on-a-Chip devices and their potential for nanobiotechnology.

17. Discuss in detail about the cell - nanostructure interaction.

18. Explain in detail about the nano-containers for drug delivery

19. Give a detail account on DNA - functionalized gold nanoparticles.

20. Give a detail account on nanobiosensor.

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RW-6108 | 503303

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

BIOELECTRONICS—II

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. Why Microfabricated devices are not generally freestanding devices ?

2. Differentiate Dry etching from Wet etching.

3. What is photometric sensor ?

4. Write a short note on DNAFET.

5. Write the lipid bilayer model of cell membrane.

6. What is the origin of Bioelectronic devices ?

7. Write a note on FET operation.

8. How to construct the crystalline membrane ?

9. Define Micromachines.

10. Write a note on HAR silicon micromachining.

Part - B (5 × 5 = 25)

Answer **all** questions.

11. (a) Discuss in detail about Chemical-Mechanical planarization.

(*Or*)

(b) Discuss in detail the fundamentals of Nanofabrication.

12. (a) Discuss in brief about design and industrial applications of biological detection technology for protein chip.

(*Or*)

(b) Write in detail about the integrated microsystem for biomolecular detection.

13. (a) Explain in detail about Lab View Programming.

(*Or*)

(b) Write notes on the similarities and variations among silicon, glass and metal electrodes using its applications.

14. (a) Explain in detail about miniaturization of electronic devices.

(*Or*)

(b) Discuss in brief about impedance monitoring.

15. (a) Illustrate the Theoretical aspects of Microfluidic.

(*Or*)

(b) Write in detail about the microfluidic components.

Part - C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail about Silicon and Cleanroom processing.

17. Discuss in detail about state-of-the-art technology for micro total analysis system.

18. Discuss in detail about hard and soft lithography.

19. Explain in detail about :

    (a) Ion-selective and

    (b) Enzymatic sensitive electrodes.

20. Discuss in detail about biofluidics theory and its applications.

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RW-6109 | 503304

M.Sc. DEGREE EXAMINATION, NOVEMBER 2010

Bioelectronics and Biosensors

BIOMEDICAL INSTRUMENTS

(CBCS–2008 onwards)

Time : 3 Hours | Maximum : 75 Marks

Part - A (10 × 2 = 20)

Answer **all** questions.

1. What are the respiration methods which are carried out in vertebrates ?

2. What are transducers and give some examples ?

3. What are amplifiers ? Give some examples.

4. What is lead system and give some recording methods ?

5. What is the pH of blood ?

6. What are ESR and GSR measurements ?

7. What are the different types of patient monitoring systems ?

8. What is MRI ?

9. What are defibrillators ?

10. What are nerve and muscle stimulators ?

Part - B (5 × 5 = 25)

Answer **all** questions choosing *either* (a) *or* (b).

11. (a) Explain the physiology of heart and lungs.

(*Or*)

(b) Explain in brief the different types of transducers.

12. (a) Briefly explain about basic components of a biomedical system.

(*Or*)

(b) Discuss about ECG, EEG, EMG, and ERG.

13. (a) Explain about cardiac output, cardiac rate, heart sound, and respiratory rate.

(*Or*)

(b) How will you measure gas volume and flow rate of CO_2, O_2 in exhaust air ?

14. (a) Explain the details about principle and working conditions of X-ray diffraction.

(*Or*)

(b) A brief discussion on computer tomography and MRI.

15. (a) Give an account of ventilators, nerve and muscle stimulators.

(*Or*)

(b) Explain Audiometers and Dializers.

Part - C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the physiology of heart and lungs, circulation and respiration.

17. Explain in details about the lead system and recording methods.

18. Explain the details of pH of blood, ESR, GSR and Plethysmography.

19. Describe about Ultrasonography, Endoscopy, Thermography.

20. Give an account of nerve and muscle stimulators, diathermy heart, lung machine audiometers and dializers.

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